

Computer Vision Project – Exercise 3

Writer Identification with VLAD Encoding and Exemplar SVMs

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1 Motivation

The goal of this exercise is to implement and evaluate a small writer identification system on the ICDAR2017 Historical Writer Identification (WI) dataset. The dataset consists of historical handwritten documents, split into a training (experiment) set and a test (benchmark) set. Each document is labeled with a writer ID.

From the lecture, we have seen that large collections of handwritten documents cannot be searched efficiently by hand. Automatic writer identification and retrieval are therefore important for applications in document analysis, digital humanities, and forensic investigations. In this exercise, I implement a bag-of-visual-words style pipeline using VLAD encoding of local descriptors, followed by optional normalization and exemplar-based classification. The system is evaluated in terms of Top-1 identification accuracy and mean average precision (mAP), as introduced in the lecture.

2 Method Overview

In this section I briefly describe the main steps of the provided `skeleton.py` code and how they relate to the pipeline from the lecture. I only give a high-level overview, without going into implementation details.

2.1 Codebook Generation (Part a)

The first step is to build a visual dictionary (codebook) from the local SIFT descriptors provided for the training set. The function `loadRandomDescriptors` randomly samples approximately 500 000 descriptors from up to 100 training files. These descriptors are then clustered using `MiniBatchKMeans` with $k = 100$ clusters in the `dictionary` function. The resulting 100 cluster centers form the codebook $\{\mu_k\}$ which is stored in `mus.pkl.gz` for later use.

2.2 VLAD Encoding (Part b)

For each image, the corresponding local descriptors are loaded and assigned to their nearest cluster center using OpenCV's `BFMatcher` and `knnMatch`. This produces a hard assignment matrix where each descriptor is associated with exactly one cluster.

The `vlad` function then computes the VLAD encoding: for each cluster k , the residuals $(x_t - \mu_k)$ of all descriptors assigned to that cluster are summed. The residual vectors for all clusters are concatenated into a single global descriptor of dimension $K \cdot D$ per image. This gives one VLAD vector per document, which serves as its global representation.

2.3 VLAD Normalization (Part c)

To mitigate visual burstiness, I implemented power normalization inside `vlad`. After computing the raw VLAD vector f , each component f_i is transformed using

$$f'_i = \text{sign}(f_i) \sqrt{|f_i|}.$$

Afterwards, the vector is ℓ_2 -normalized using `sklearn.preprocessing.normalize`. The script evaluates the VLAD encodings *without* power normalization and with power normalization in order to compare their influence on Top-1 accuracy and mAP.

2.4 Exemplar SVMs (Part d)

In the final step, exemplar classification is applied to further improve the representation. For each test encoding, an individual linear SVM is trained using that single test VLAD vector as positive sample and all training VLAD vectors as negatives. This is done in the `esvm` function using `LinearSVC` with $C = 1000$ and `class_weight='balanced'`. The learned weight vector w is then ℓ_2 -normalized and used as a new global descriptor for that test image.

In this way, each test sample is mapped to a new feature space where the direction of w emphasizes the differences between that test instance and the background training set. The pairwise distances between these new descriptors are then used for retrieval and writer identification.

3 Usage and Arguments

The script `skeleton.py` is executed from the command line. A typical call that I used is:

```
python3 exercise3.py \
  --in_train icdar17_local_features/train \
  --labels_train icdar17_local_features/icdar17_labels_train.txt \
  --in_test icdar17_local_features/test \
  --labels_test icdar17_local_features/icdar17_labels_test.txt \
  --suffix _SIFT_patch_pr.pkl.gz \
  --powernorm \
  --overwrite \
  --C 1000
```

The most important arguments are:

- `--in_train`: path to the folder containing training feature files.

- `--labels_train`: text file listing training file names and their writer labels.
- `--in_test`: path to the folder containing test feature files.
- `--labels_test`: text file with test file names and labels.
- `--suffix`: suffix of the feature files to load (here: `_SIFT_patch_pr.pkl.gz`).
- `--overwrite`: if set, recompute encodings even if cached files exist.
- `--powernorm`: if set, apply power normalization to the VLAD encodings.
- `--gmp`: (not used in my experiments) would enable generalized max pooling.
- `--gamma`: GMP regularization parameter (only relevant with `--gmp`).
- `--C`: SVM regularization parameter C for the exemplar SVMs.

The script automatically performs the following sequence: build or load the codebook, encode all test images with VLAD, evaluate VLAD without and with power normalization, compute VLAD encodings for the training set, train exemplar SVMs for all test samples, and re-evaluate using the new descriptors.

4 Results

All experiments were run on the provided ICDAR2017 historical local feature set using $K = 100$ clusters and the SIFT patch descriptors. I report Top-1 identification accuracy and mAP as printed by the script.

VLAD Without Power Normalization

For standard VLAD encoding (sum pooling, ℓ_2 -normalization, but *no* power normalization) on the test set, I obtained:

- Top-1 accuracy: 0.8086
- mAP: 0.6152

VLAD With Power Normalization

When enabling power normalization in addition to ℓ_2 -normalization, the performance improved slightly:

- Top-1 accuracy: 0.8233
- mAP: 0.6311

VLAD + Exemplar SVMs

After computing VLAD encodings for the training set and applying exemplar SVMs to obtain new descriptors for the test set, the evaluation further improved:

- Top-1 accuracy: 0.8889
- mAP: 0.7533

These numbers show a clear gain from both power normalization and exemplar classification.

5 Discussion

The experiments confirm several points discussed in the lecture and in the exercise sheet:

- **Effect of Power Normalization:** Power normalization is designed to reduce the influence of bursty visual patterns in VLAD. In my results, enabling power normalization gave a consistent improvement in both Top-1 accuracy and mAP. The gain in mAP from 0.6152 to 0.6311 suggests that the ranking of relevant documents becomes slightly more stable when frequent residuals are down-weighted.
- **Effect of Exemplar SVMs:** The largest improvement came from the exemplar SVM step. By training an individual SVM per test sample against the training encodings, the model effectively re-embeds each test document into a space tuned to separate it from the background set. This leads to a more discriminative representation and boosts mAP from roughly 0.63 to 0.75, and Top-1 accuracy from about 0.82 to almost 0.89. This confirms that exemplar classification is a powerful way to refine global descriptors for retrieval.
- **Behaviour on Historical Data:** The ICDAR2017 dataset consists of challenging historical documents with noise and degradation. Despite this, the bag-of-visual-words/VLAD approach with pre-computed SIFT patches already achieves a reasonably strong baseline. The improvements from normalization and exemplar SVMs indicate that careful post-processing of global descriptors can compensate to some degree for the variability in the data.
- **Computational Aspects:** The exemplar SVM step is computationally more expensive than plain VLAD evaluation, since it trains one SVM per test instance. In my run, training 3600 exemplar SVMs took several minutes. For larger datasets, this could become a bottleneck, but the accuracy gain might justify the cost in applications where retrieval quality is critical.

6 Conclusion

In this exercise I implemented a writer identification pipeline based on VLAD encoding of local SIFT descriptors and evaluated it on the ICDAR2017 Historical WI dataset. The basic VLAD representation without power normalization already reached a Top-1 accuracy of about 0.81 and an mAP of about 0.62.

Adding power normalization on top of VLAD led to a modest but consistent improvement in both metrics, highlighting the importance of handling visual burstiness. The most significant improvement came from incorporating exemplar SVMs, which yielded a Top-1 accuracy close to 0.89 and an mAP of around 0.75. This shows that exemplar-based re-embedding of global descriptors can substantially enhance retrieval and identification performance.

Overall, the exercise helped me to connect the theoretical concepts from the lecture (VLAD encoding, normalization, exemplar classification) with a practical implementation on a real historical dataset, and to observe their impact on writer identification performance in practice.